Last updated in Aug 2026

Martín Cogo Belver

Computer Science Degree

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Profile

Computer Science graduate with focus on Artificial Intelligence and Computer Vision problem solving. Proven experience designing and implementing real-time inference systems over video streams, as well as training deep learning models for computer vision applications.

Experience

MindColab, Computer Scientist – July 2025 – present

Designed and implemented a real-time inference engine built on GStreamer pipelines, capable of processing up to 6 parallel RTSP streams at 720p and 25 FPS, maintaining frame drop below 1% under load. The system follows a multi-process architecture with shared-memory IPC to minimize unnecessary copies and reduce end-to-end latency.

- **Flow control**: Implemented backpressure handling using GStreamer queues to sustain stable throughput under concurrent load.
- **Acceleration stack**: GPU-based implementation leveraging NVIDIA DeepStream, NVMM buffers, CUDA kernels (Numba), and CUDA-enabled OpenCV for preprocessing.
- **Inference optimization**: Executed YOLO/PyTorch models with frames kept predominantly on GPU, reducing CPU↔GPU transfer overhead.
- **Lifecycle & operations**: Supported controlled restarts, hot configuration reloads via a custom DSL, and graceful shutdowns in 24/7 environments.
- **Observability**: Measured per-stage pipeline latency and monitored GPU, VRAM, RAM, and CPU utilization under sustained load.
- **Technical impact**: Migrated from a CPU-bound solution to GPU acceleration, enabling real-time stream retransmission where it was previously not feasible.

Municipality of Godoy Cruz, Web Developer – June 2025 – present

Maintain and extend a legacy GeneXus-generated Java web application for municipal debt and tax payments, including refactoring integrations with multiple banking payment gateways, and building an internal LLM-based chat interface for semantic querying of municipal regulations.

- **Legacy maintenance**: Maintain and resolve tickets on a GeneXus-generated Java codebase, within the municipal debt and tax payment module.
- **Payment integrations**: Refactored integrations with multiple banking payment gateway APIs (Banco Macro, Mercado Pago, ePagos, XNet, Pago Fácil, Supervielle) in the transaction and amount-submission flow.
- **LLM chat interface**: Built an internal web chat interface for semantic querying of municipal regulations via an LLM.
- View Project

Klari Inc., Data Scientist – Jan 2025 – Feb 2025

Designed and implemented a semantic analysis system for conversations generated by an LLM-based chatbot, aimed at extracting thematic patterns and producing actionable reports on user behavior and interaction quality.

- **NLP pipeline**: Semantic representation of conversations using embeddings.
- **Thematic analysis**: Unsupervised clustering to extract conversational patterns.

CICDa Research Group, Thesis Research – Jan 2024 – Jan 2025

Conducted publication-oriented research focused on satellite image segmentation for post-disaster building damage detection and estimation, using temporal image comparison and Siamese deep learning architectures.

- **xBD dataset**: Trained on ~2,400 VHR satellite images for Building Damage Assessment tasks.
- **End-to-end pipeline**: Preprocessing, training, inference, and post-processing for semantic segmentation and multitemporal structural change detection.
- **Computer vision modeling**: Trained deep learning models for post-disaster building semantic segmentation.
- **Experimental evaluation**: Quantitative analysis using classification and segmentation metrics (Precision, Recall, F1-score).
- **Compute**: Training executed on a High Performance Computing cluster with NVIDIA Tesla V100 GPUs.
- View Project

Technical Skills

Programming Languages: Python, Java, TypeScript, C++, Rust

AI & ML: PyTorch, TensorFlow, Scikit-learn, Ultralytics, Anomalib

Vision & Data: OpenCV, CUDA, Numba, GStreamer, Pandas, NumPy

Backend & Web: FastAPI, Java, Tomcat, GeneXus, Nginx, React, Tailwind, Bun

Databases: SQLite, Redis, MongoDB

Tools: Docker, Linux, Git

Education

National University of Cuyo, Bachelor of Computer Science 2020 – 2024

Achievements and Languages

- Highest GPA (8.7 on a ten-point scale) in the Computer Science program, Class of 2024.
- Ranked among the top 15 engineering students at UNCUYO (2024), based on academic GPA.
- English: Upper-Intermediate (B2), Cambridge certified.